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WaveSP-Net

Learnable Wavelet-Domain Sparse Prompt Tuning for Speech Deepfake Detection

基于可学习小波域稀疏提示调优的语音深度伪造检测

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Background: Speech Deepfake Detection

What is SDD?

Identifying artificially generated or manipulated speech audio, distinguishing it from bonafide human speech.

Key Challenges

- Cross-domain generalization to unseen attacks
- Diverse speech codecs & compression formats
- Computational cost of large SSL models
- Overfitting with data-driven approaches

Attack Types

- Speech Synthesis
- Voice Conversion
- Voice Cloning
- Audio Manipulation

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Background: Front-end Design for SDD

DSP-based Front-ends

- Fixed transforms
  (短时傅立叶变换 STFT,
  线性频率倒谱系数LFCC,
  常数Q变换CQT)
- Lightweight & interpretable

Limited representation capacity
Cannot adapt to new domains

SSL-based Front-ends

- XLSR, Wav2Vec 2.0
- Better detection performance

Computationally demanding
Millions of parameters
Prone to overfitting

Our Approach: DSP transforms + SSL via Prompt Tuning

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Background: Prompt Tuning for PEFT

Why Prompt Tuning?

Adapt a speech foundation model to new domains while keeping the number of trainable parameters small.

1. Freeze

Keep XLSR model parameters frozen

2. Prepend

Add learnable prompt tokens to transformer layers

3. Optimize

Update only prompt parameters (tiny params)

Only 1.298% of total parameters are trainable

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Motivation: Visual Fourier Prompt Tuning (NeurIPS '24)

Zeng et al., "Visual Fourier Prompt Tuning", NeurIPS 2024 [arxiv: 2411.01327]

Key Idea of VFPT

- Incorporates 2D FFT into prompt embeddings
- Integrates both spatial & frequency domain info
- Inspired by human visual cognition
- Partial prompts → FFT → real component → concat
- Only 0.57~0.66% model parameters tuned

VFPT Results on VTAB-1k:

73.20% mean accuracy (vs. 69.43% VPT-Deep)
Outperforms full fine-tuning in 22/24 tasks

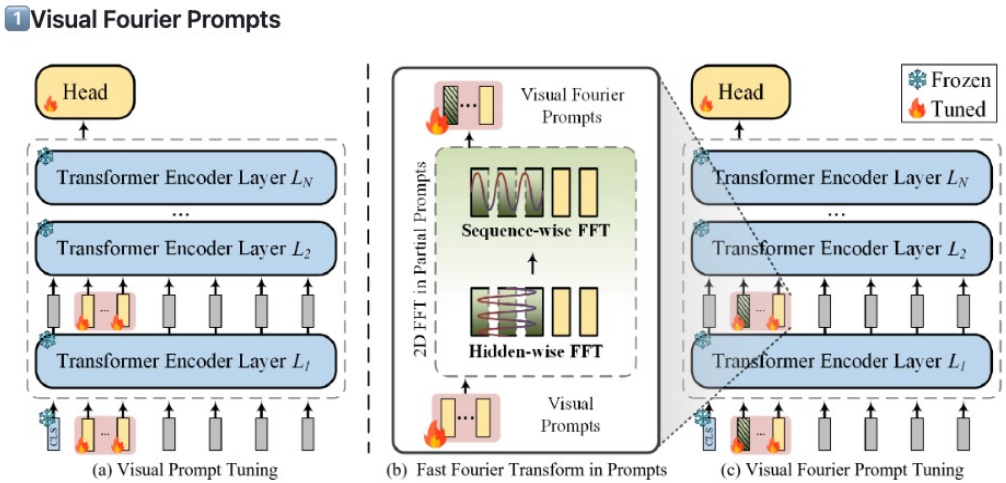

Fig. 1(a-b): PT vs FourierPT-XLSR (adapted from VFPT for speech)

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03

Proposed Methods

FourierPT-XLSR · WSPT-XLSR · Partial-WSPT-XLSR

Architecture Overview (WaveSP-Net)

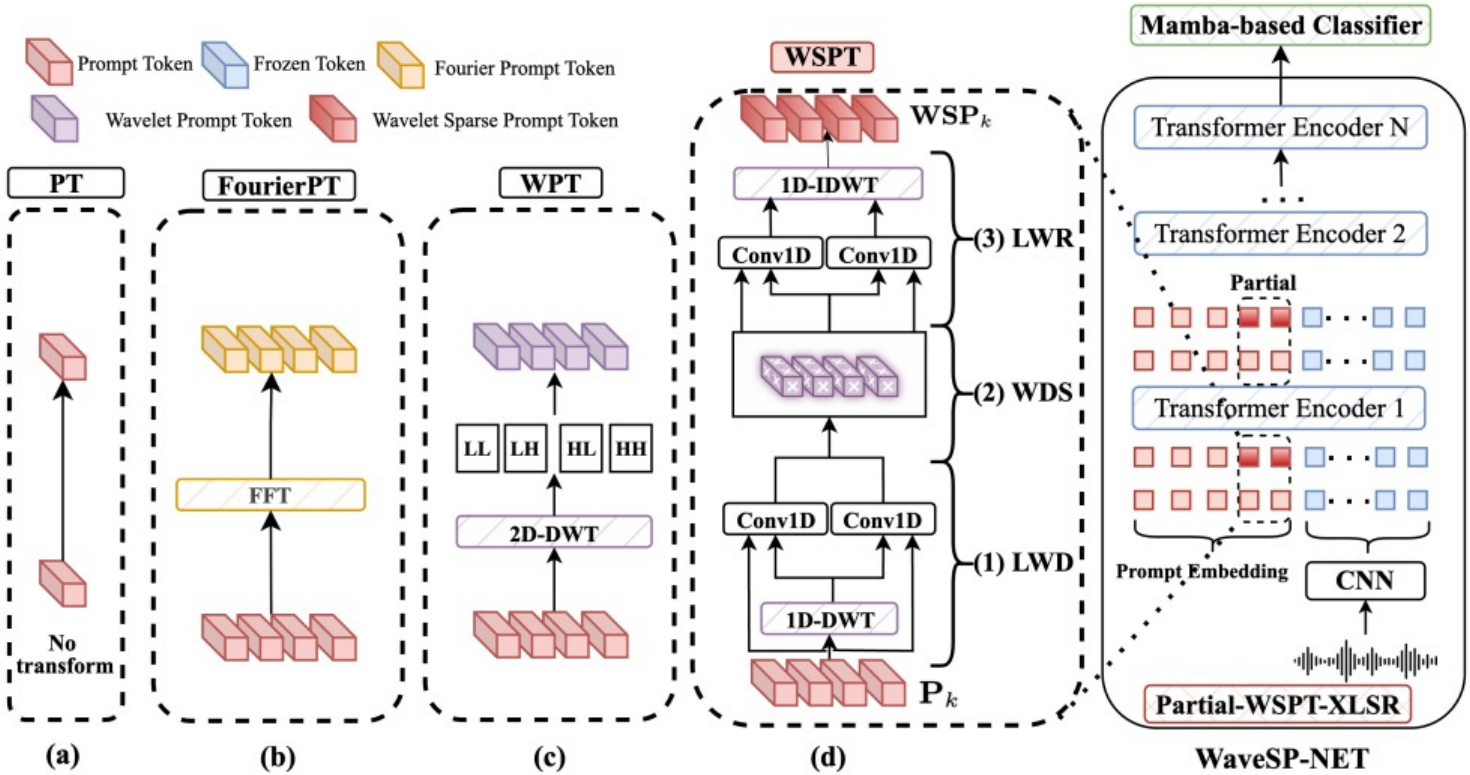


Fig. 1. Overview of the WaveSP-Net architecture. The figure illustrates five different XLSR-based front-end variants: (a) PT-XLSR, (b) FourierPT-XLSR, (c) WPT-XLSR, (d) WSPT-XLSR and Partial-WSPT-XLSR. The proposed WaveSP-Net (rightmost panel) integrates a Partial-WSPT-XLSR front-end (bottom right) with a Mamba-based classifier (top right). (FFT: Fast Fourier Transform; DWT: Discrete Wavelet Transform; IDWT: Inverse Discrete Wavelet Transform)

Method 1: FourierPT-XLSR

Adapted from VFPT (NeurIPS 2024) for speech domain

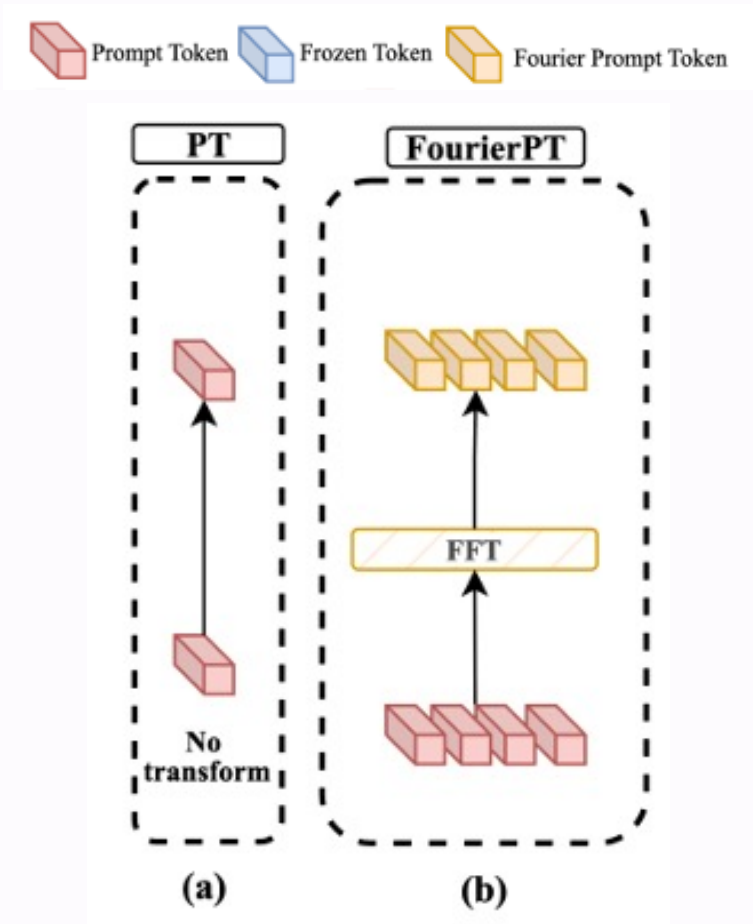
- Augments FFT features into prompt embeddings
- Applied to frozen XLSR for deepfake detection

Limitation:

FFT uses non-localized sine/cosine bases

→ less effective for abrupt signal changes in speech

This motivates us to explore Wavelet Transform



Method 2: WSPT-XLSR — Wavelet Enhancement

Key Idea: Use DWT to enhance prompt embeddings with multi-resolution features

Why Wavelet over Fourier Transform?

- Joint time-frequency localization with adaptive resolution
- Robust to signals with abrupt changes
- Separates coarse structure (low-pass) from details (high-pass)

Three-Step Pipeline

(1) LWD

Learnable Wavelet Decomposition
→ Low-pass F_0 + High-pass F_1
→ Filters are learned

→

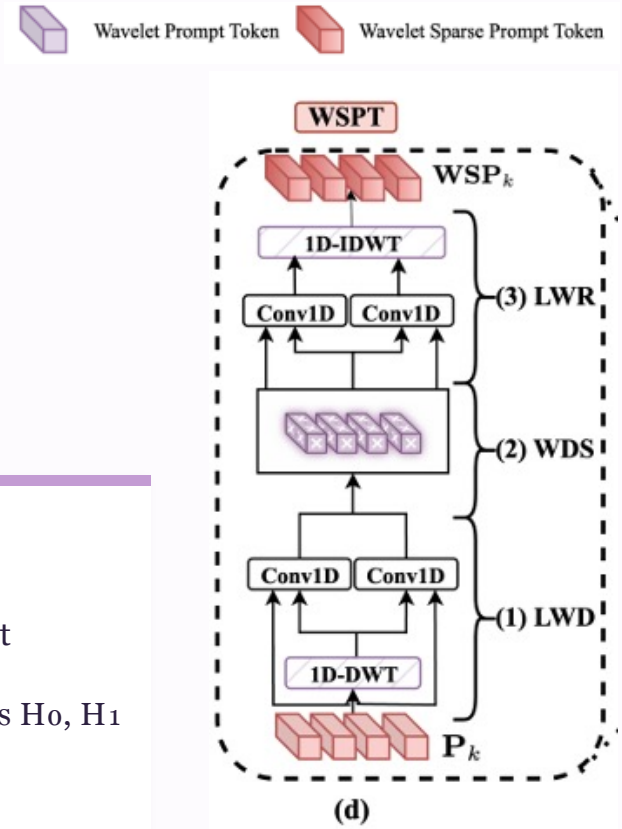
(2) WDS

Wavelet Domain Sparsification
→ Random selection (ρ=0.1)
→ Implicit regularizer

→

(3) LWR

Learnable Wavelet Reconstruction
→ Synthesis filters H_0, H_1
→ Jointly learned



Method 3: Partial-WSPT-XLSR

Selective Wavelet Enhancement

- **Only partial prompt tokens undergo wavelet transform**

From p = 10 total prompt tokens per layer:

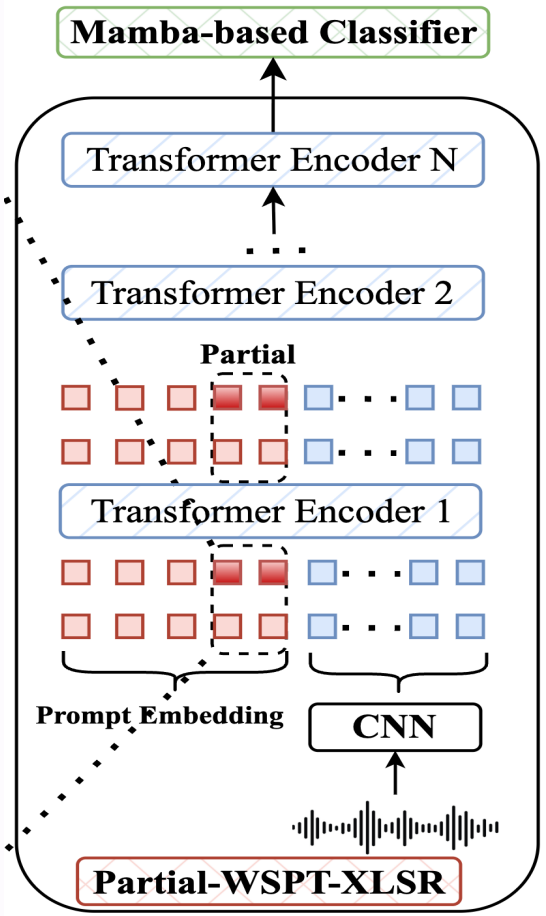
· m = 4 wavelet sparse prompt tokens

· 6 regular (untransformed) tokens

2.3. WaveSP-Net

After computing $\mathbf{WSP}_k$, we obtain the final prompt representation that integrates both enhanced and untransformed tokens:

$$\tilde{\mathbf{P}}_k = [\mathbf{P}_k^{(1:p-m)}, \mathbf{WSP}_k] \in \mathbb{R}^{p \times d}, \quad (1)$$



WaveSP-NET

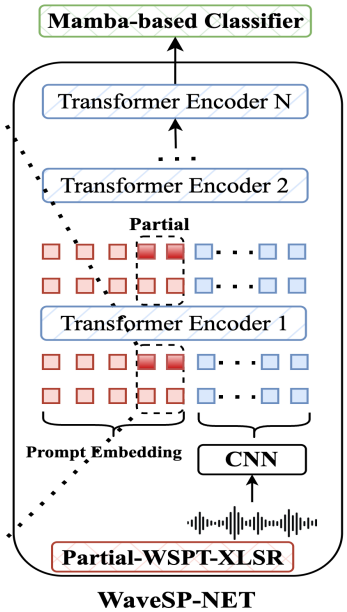
WaveSP-Net: Full Architecture

Partial-WSPT-XLSR Front-end + Bidirectional Mamba Back-end

Front-end: Partial-WSPT-XLSR

- Frozen XLSR-300M backbone
- Learnable wavelet filters (LWD→WDS→LWR)
- Partial prompt token enhancement
- Input: 16kHz, 4s → (201, 1024)

$$[\mathbf{Z}_k, \mathbf{E}_k] = L_k([\tilde{\mathbf{P}}_k, \mathbf{E}_{k-1}]), \quad k = 1, 2, \ldots, \ell$$



Back-end: Mamba Classifier

- Bidirectional Mamba architecture
- 12 Mamba-based blocks
- Long-range temporal dependencies
- Linear time complexity

Training: Only prompt embeddings + wavelet filters + Mamba classifier updated (XLSR frozen)

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Experimental Setup Results

Datasets & Implementation

Deepfake-Eval-2024 (DE24)

- 88 web domains, 42 languages
- Varying acoustic conditions
- Strictly unseen attacks
- 4-second segment chunking

SpoofCeleb

- Large-scale benchmark
- Train: A01-A10, Eval: A15-A23
- In-the-wild conditions

Implementation Details

Parameter	Value	Parameter	Value
GPU	Tesla V100	Batch size	16
Audio	16kHz, 4s	Learning rate	5×10^{-4}
Prompt tokens	$p = 10$	Optimizer	Adam
Sparsity ratio	$\rho = 0.1$	Loss	Cross-entropy
Mamba blocks	12	Epochs	Up to 100

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Results: Three Novel Front-ends Comparison

Deepfake-Eval-2024

Model	EER (%) ↓	ACC ↑	F1 ↑	AUC ↑
FourierPT-XLSR	16.58	83.42	79.53	90.35
WSPT-XLSR	13.15	86.85	83.84	93.33
Partial-WSPT-XLSR	**10.58**	**89.42**	**86.35**	**94.26**

SpoofCeleb

Model	EER (%) ↓	ACC ↑	F1 ↑	AUC ↑
FourierPT-XLSR	0.23	99.84	99.87	99.86
WSPT-XLSR	0.19	99.89	99.92	99.91
Partial-WSPT-XLSR	**0.13**	**99.87**	**99.93**	**99.99**

Wavelet-based methods consistently outperform Fourier. Partial-WSPT-XLSR achieves best results on both datasets.

Results: SOTA Comparison on DE24

Model	Params	EER ↓	ACC ↑	F1 ↑	AUC ↑
AASIST	0.3M	16.99	83.60	77.80	90.60
RawNet2	18M	20.91	81.70	86.00	87.60
P3	317M	15.38	85.50	81.00	92.00
XLS-R-1B	965M	11.85	86.83	89.43	94.35
BCM	-	-	89.00	87.00	93.00
PT-XLSR	4.145M	20.40	79.60	77.19	90.21
WPT-XLSR	4.145M	14.39	85.61	81.01	91.29
WaveSP-Net	**4.146M (1.3%)**	**10.58**	**89.42**	**86.35**	**94.26**

10.58%

EER (Best)

10.72%

Relative Improvement

1.298%

Trainable Params

Results: SOTA Comparison on SpoofCeleb

Model	Params	EER ↓	ACC ↑	F1 ↑	AUC ↑
AASIST	0.3M	2.37	71.38	81.25	83.56
RawNet2	18M	1.12	87.23	88.92	92.14
PT-XLSR	4.145M	0.26	99.74	99.85	99.93
WPT-XLSR	4.145M	0.15	99.85	99.92	99.97
WaveSP-Net	**4.146M (1.3%)**	**0.13**	**99.87**	**99.93**	**99.99**

0.13% EER

99.87% ACC

99.93% F1

99.99% AUC

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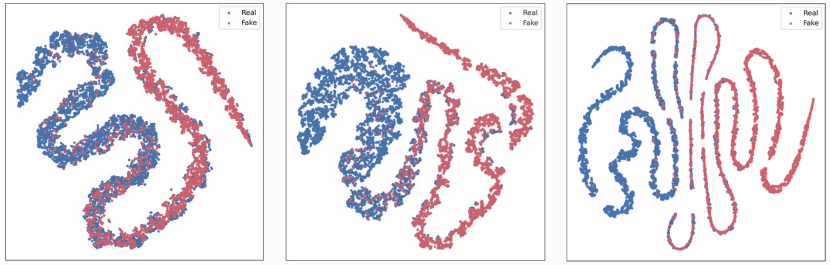
Ablation Study & Visualization

Ablation on DE24

Config	EER (%) ↓	ACC ↑
WaveSP-Net (full)	**10.58**	**89.42**
w/o LWD	12.97	87.03
w/o WDS	14.34 (+35.5%)	85.66
w/o LWR	11.33	88.67
Fixed Filters	16.55 (+56.4%)	83.45
Sparsity ρ=0.5	12.42	87.58
Sparsity ρ=0.7	13.84	86.16
WSP tokens=2	11.23	88.77
WSP tokens=6	14.86	85.14

t-SNE Visualization (DE24)



(a) FourierPT (b) WSPT (c) Partial-WSPT

Key Findings

- WDS removal → largest EER drop (35.5%)
- Fixed filters → 56.4% EER increase
- Optimal: ρ=0.1, 4 WSP tokens
- Partial-WSPT shows clear cluster separation

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Conclusion

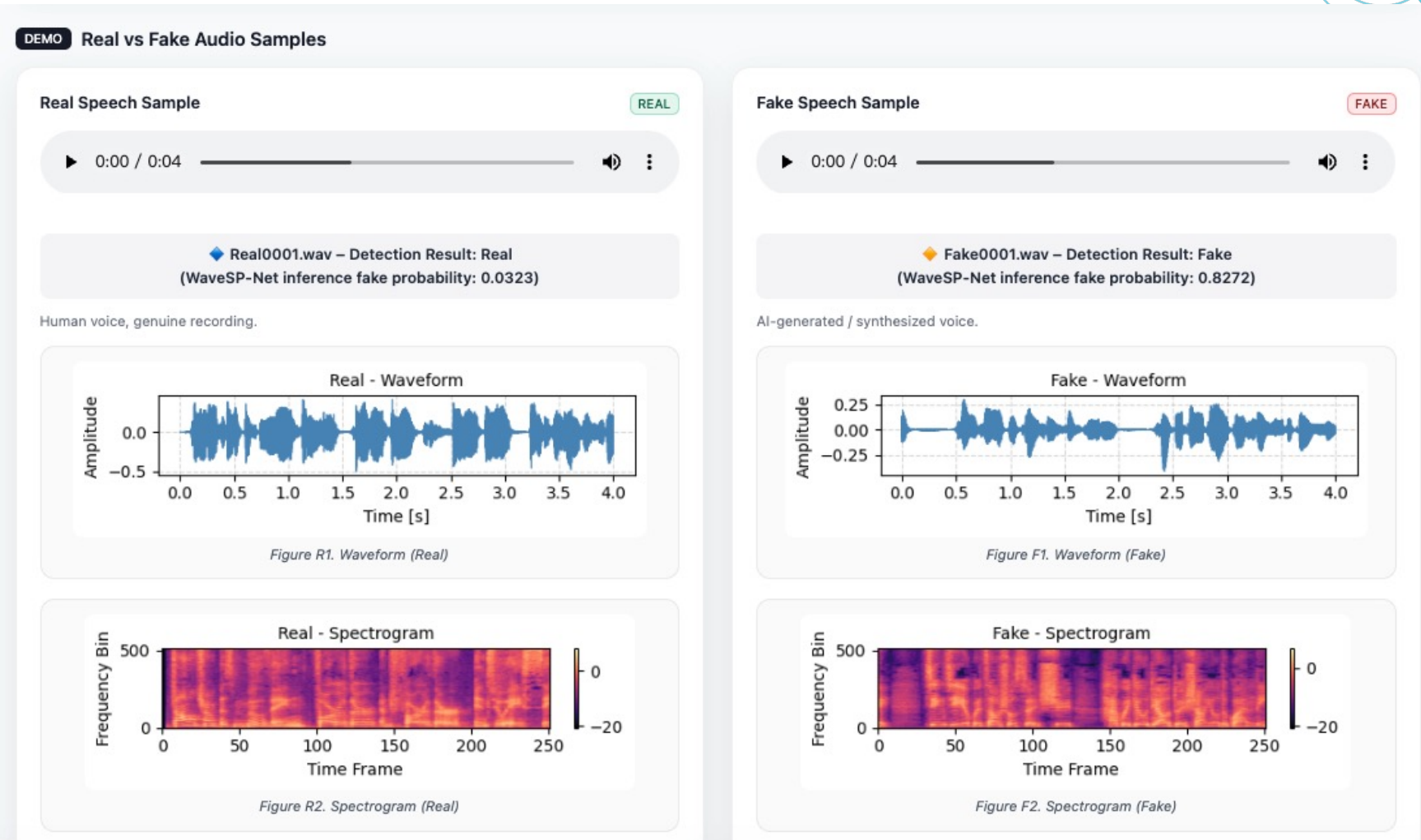
Novel integration of learnable wavelet transforms into prompt tuning for SDD

Only 1.298% trainable parameters with frozen XLSR backbone

SOTA results on Deepfake-Eval-2024 and SpoofCeleb benchmarks

Classical signal processing + modern deep learning = better generalization

Demo Website



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Thank you for listening !